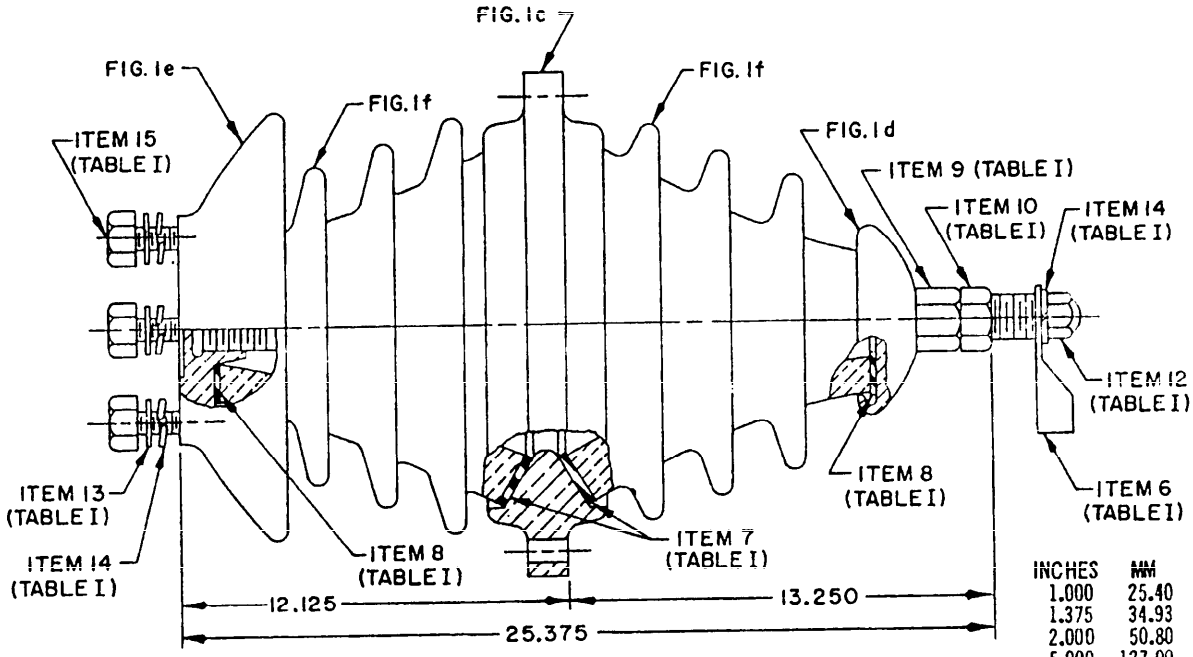
MILITARY SPECIFICATION SHEET

INSULATORS, FEEDTHRU
(STYLE 31)

The complete requirements for procuring the insulators described herein shall consist of this document and the latest issue of Specification MIL-I-23264.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
1.000	25.40
1.375	34.93
2.000	50.80
5.000	127.00
12.125	307.98
13.250	336.55
25.375	644.53
26.875	682.63

FIGURE 1a. Feedthru insulator assembly.

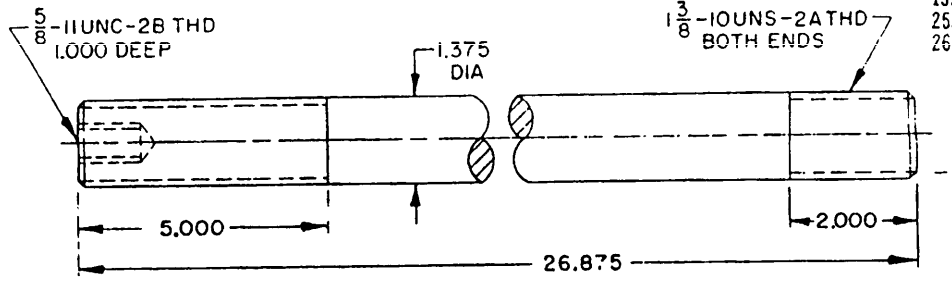


FIGURE 1b. Anchor rod, (Item 11, Table I)

FSC 5970

FIGURE 1. Style 31 feedthru insulator.

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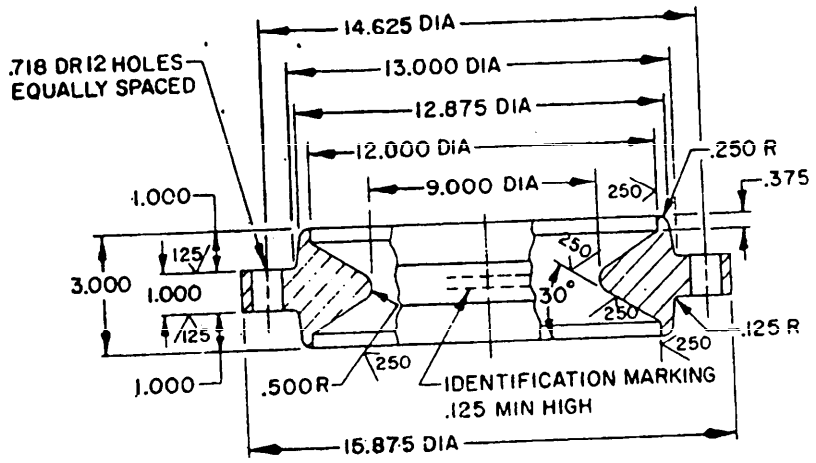
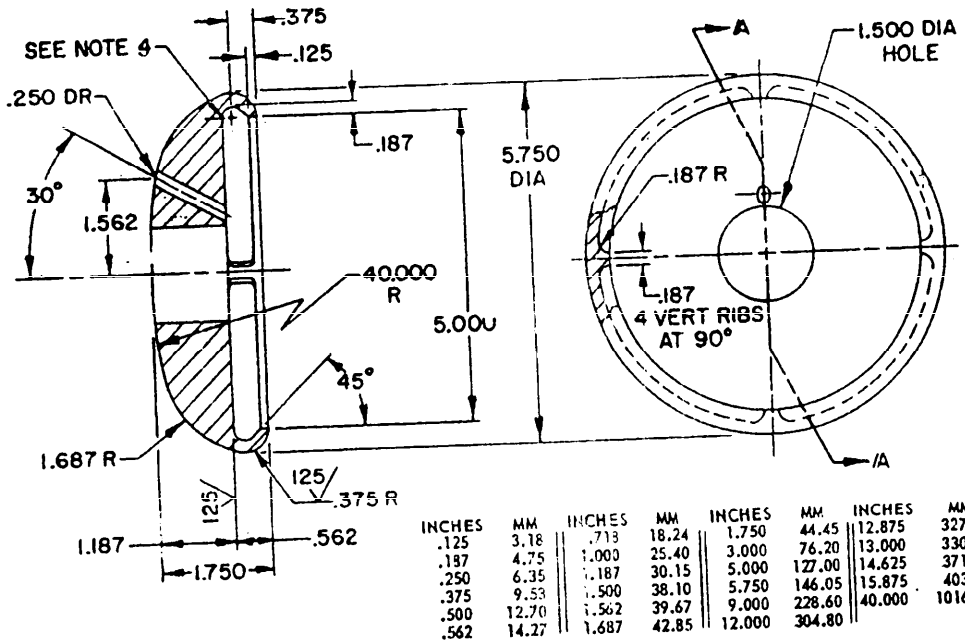


FIGURE 1c. Flange.
(Item 5, Table I)



INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.125	3.18	.718	18.24	1.750	44.45	12.875	327.03
.187	4.75	1.000	25.40	3.000	76.20	13.000	330.20
.250	6.35	1.187	30.15	5.000	127.00	14.625	371.48
.375	9.53	1.500	38.10	5.750	146.05	15.875	403.23
.500	12.70	1.562	39.67	9.000	228.60	40.000	1016.00
.562	14.27	1.687	42.85	12.000	304.80		

FIGURE 1d. Lower cap.
(Item 4, Table I)

FIGURE 1. Style 31 feedthru insulator. - Continued.

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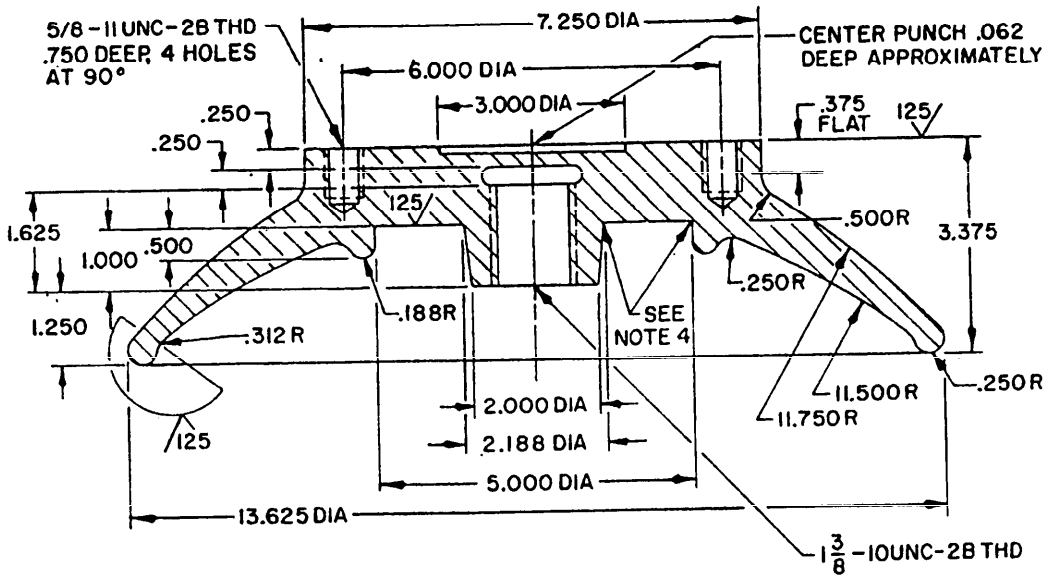
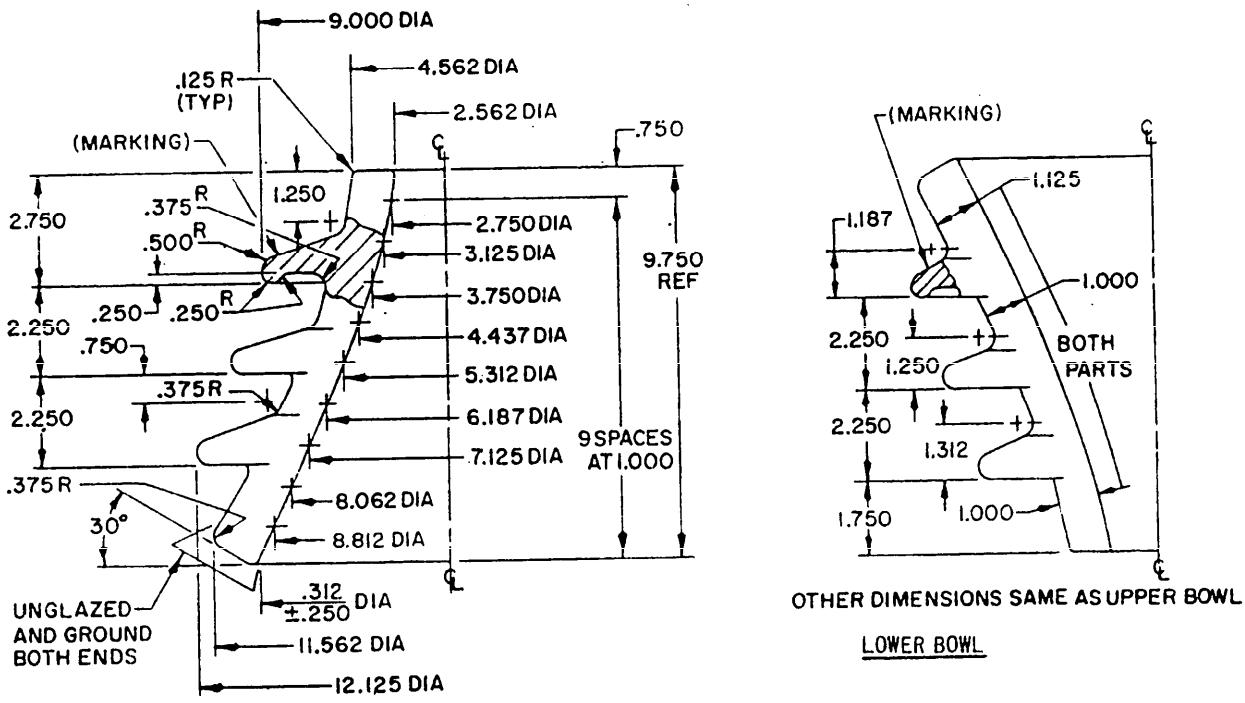


FIGURE 1e. Upper cap.
(Item 3, Table 1)



UPPER BOWL

LOWER BOWL

FIGURE 1f. Insulator bowl.
(Items 1 & 2, Table 1)

INCHES	MM	INCHES	MM	INCHES	MM
.062	1.57	1.625	41.28	5.000	127.00
.125	3.18	1.750	44.45	5.312	134.92
.188	4.78	2.000	50.80	6.000	152.40
.250	6.35	2.188	55.58	6.187	157.15
.312	7.92	2.250	57.15	7.125	180.98
.375	9.53	2.562	65.07	7.250	184.15
.500	12.70	2.750	69.85	9.000	228.60
.750	19.05	3.000	76.20	9.750	247.65
1.000	25.40	3.125	79.38	11.500	292.10
1.125	28.58	3.375	85.73	11.562	293.67
1.187	30.15	3.750	95.25	11.750	298.45
1.250	31.75	4.437	112.70	12.125	307.98
1.312	33.32	4.562	115.87	13.625	346.08

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Prior to shipment, the hexagon nut shall be tightened with sufficient pressure to keep insulator bowls intact. The pressure applied shall not cause serious deformation to the gaskets.
4. Machine finish corner square.

FIGURE 1. Style 31 feedthru insulator. - Continued.

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TABLE I. Item description.

Item No.	Description	Material	No. per assembly
1	Insulator bowl, upper	Ceramic, grade 222 or better, per MIL-I-10	1
2	Insulator bowl, lower		1
3	Cap, upper	Bronze per QQ-B-726	1
4	Cap, lower		1
5	Flange		1
6	Terminal lug, Amp Inc. #35669 or equal	Copper, hot electro tinned	1
7	Gasket, 12.000 OD, 9.125 ID, 0.125 THK, ±.005	Rubber, type 1, class 2, per MIL-G-1149	2
8	Washer, 4.937 OD, 2.250 ID, 0.125 THK, ±.005	Copper, soft annealed per QQ-C-576	2
9	Hexagon nut, 1-3/8-10UNS-2B	Brass per QQ-B-637; cadmium plated, type II, class C, per QQ-P-416	1
10	Hexagon jam nut, 1-3/8-10UNS-2B		1
11	Anchor rod	Steel, Alloy #2, per MIL-S-890	1
12	Hexagon head cap screw, 5/8-11UNC-2A, 1 in. long, per MS35309-158	Cadmium plated, type II, class C, per QQ-P-416	1
13	Washer per MS15795-820		4
14	Lockwasher per MS35338-145 or -164		5
15	Hexagon head cap screw, 5/8-11UNC-2A, 1-1/2 in. long, per MS35307-462		4

TABLE II. Type designation.

Type designation	Dimensions
NL222B31-203	See figure 1

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REQUIREMENTS:

Dimensions: See figure 1 and table I.

Tolerances:
Ceramic - Unless otherwise specified, the tolerances for ceramic parts shall be as specified in table IV of MIL-I-23264.
Metal - Unless otherwise specified, the tolerance for metal parts shall be ±.020 inch. Tolerance on angles shall be ±1°.

Component parts:
Ceramic parts - Ceramic per MIL-I-10 shall be grade 222 or better.
Metal parts - Unless otherwise specified, all bearing and exposed metal surfaces shall be machine finished to 125/.

Cantilever: Insulators shall have the 1-3/8-inch diameter hexagon nut tightened with sufficient pressure to elongate the anchor rod 0.018 inch. The jam nut shall then be tightened and the insulator flange shall be bolted to a rigid steel frame. An ultimate cantilever load of 100,000 inch-pounds, minimum, shall be applied at 90 degrees to the axis of the insulator and at the end of a 100-inch lever arm.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI, MU
Navy - SH, EC
Air Force - 85
DSA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force -

Preparing activity:
Navy - EC

(Project 5970-0159)

Revision letters are not used to indicate changes due to the extensiveness of the changes.

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